

VK Innovative Technologies

Project 17 - LCD 16x2 Display Basics: Full Working Arduino Code

Use this code after wiring the LCD exactly as shown in the guide image. This version keeps the working code inside loop(), as requested.

LCD 16x2 Pin	Connect To
VSS	GND
VDD	5V
V0	10K potentiometer middle pin
RS	D12
RW	GND
EN	D11
D4	D5
D5	D4
D6	D3
D7	D2
A	5V via 220 ohm resistor
K	GND

Main Working Code

```
#include <LiquidCrystal.h>

// LCD pin order: RS, EN, D4, D5, D6, D7
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

void setup()
{
    // Start the LCD with 16 columns and 2 rows
    lcd.begin(16, 2);
}

void loop()
{
    // Print first line
    lcd.setCursor(0, 0);
    lcd.print("VK INNOVATIVE  ");

    // Print second line
    lcd.setCursor(0, 1);
    lcd.print("TECHNOLOGIES  ");

    // Keep the message visible and refresh every second
    delay(1000);
}
```

Expected LCD Output

```
VK INNOVATIVE
TECHNOLOGIES
```

Important Notes

- If the LCD shows black boxes, rotate the 10K potentiometer slowly.
- If the LCD is blank, check 5V, GND and backlight pins A/K.
- If random characters appear, check RS, EN and D4-D7 connections.
- The extra spaces after the printed text clear leftover characters on the LCD row.

Optional Static Display Code

This version prints the message once in setup(). Use it only if you are comfortable with an empty loop.

```
// Optional version: one-time display in setup()
#include <LiquidCrystal.h>

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

void setup()
{
    lcd.begin(16, 2);
    lcd.setCursor(0, 0);
    lcd.print("VK INNOVATIVE");
    lcd.setCursor(0, 1);
    lcd.print("TECHNOLOGIES");
}

void loop()
{
    // Nothing required here because the message is static
}
```

Quick Upload Steps

1. Open Arduino IDE.
2. Select Board: Arduino Nano.
3. Select Processor: ATmega328P.
4. Select correct COM port.
5. Paste the main working code.
6. Click Upload.
7. Adjust LCD contrast after upload.